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Great Observatories: Vera Rubin Observatory

The Legacy Survey of the Sky

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Introduction

Most telescopes are built to look at things. The observatory in this volume is built to look at everything. On a mountaintop in northern Chile, an eight-and-a-half-meter telescope now sweeps the entire southern sky every few nights, photographing it over and over for ten years. Each exposure blankets a swath of sky dozens of full Moons wide, and the instrument carries the largest digital camera ever constructed, and moves between targets almost faster than a person can register the motion. Its product is not a picture of a chosen object but a continuous, decade-long movie of the whole visible universe - galaxies, stars, asteroids, explosions, and anything else that changes or sits still. When the survey is complete, the sky will have been mapped more deeply than any earlier optical survey, and the archive will hold records of tens of billions of objects.

The telescope began its working life with two names, one for the machine and one for the mission. The machine is the Simonyi Survey Telescope, standing on Cerro Pachon in the Chilean Andes. The mission is the Legacy Survey of Space and Time, a ten-year program run jointly by the United States National Science Foundation and Department of Energy. The observatory as a whole carries a third name, the one the world knows it by: the Vera C. Rubin Observatory, honoring the American astronomer whose measurements of how galaxies rotate provided the first persuasive evidence that most of the matter in the universe cannot be seen. The choice was deliberate and resonant. Rubin spent her career showing that the universe holds far more than the eye can inventory, and the observatory that bears her name was built to inventory it anyway - every moving, flaring, and fading thing, down to the limit of its enormous camera.

This book tells the story of that machine and its mission. It is a history with a surprisingly long fuse: the idea of a telescope that would survey the whole sky digitally was sketched by astronomers in the 1990s, argued over in committee reports for a decade, funded as a joint venture of two government agencies with very different scientific cultures, and assembled across a quarter century of work - a mirror cast in a rotating oven under a football stadium in Arizona, a camera the size of a small car built in a California accelerator laboratory, and a mountaintop facility in Chile whose construction began with shovels in 2015 and ended with first images in June 2025. The survey itself began in late June 2026 and is now under way.

The chapters ahead explain what makes the design unusual - a telescope with three mirrors carved from two pieces of glass, a survey strategy whose decade-long observing plan was forged in thousands of simulated skies, a field of view that had never been achieved at that aperture, and an attitude of relentless motion rather than patient staring. They cover the science the survey was designed for: dark matter and dark energy, the history written in billions of galaxies, the time-domain universe of supernovae and stellar flickers, and a census of the solar system's small bodies that will multiply our inventory of asteroids and comets. They follow the engineering: the alert stream that announces changes in the sky within a minute of their detection, and the data pipelines that turn twenty terabytes a night into usable knowledge. And they take the story to the present: the commissioning discoveries, the first alerts, the citizen projects, and the questions a ten-year movie of the universe is expected to answer - or raise.

This book follows the series' habit of taking instruments seriously as characters. Telescopes are usually described by their numbers and their discoveries; here, as in the companion volumes, the machine's personality gets the same attention as its specifications - how it was imagined, who fought for it, how it nearly stumbled, and what its existence does to the practice of science around it. The Rubin Observatory is the youngest instrument in this series, and the first one whose story is still being written while the ink is drying, which gives these pages a slightly different tension than the histories of venerable telescopes: this is not a retrospective but a birth announcement. No specialist knowledge is needed. Wherever the survey touches something technical, the text explains the idea in plain language: what a wide-fast-deep survey is, why a mirror can have two faces, how a camera weighing three tons rides a telescope that never stops, and why an observatory named for the scientist who proved we cannot see most of the universe may be the right instrument to find out what the invisible part is. The story of the Vera C. Rubin Observatory is the story of a new kind of telescope - one that does not choose what to observe, because its purpose is to make sure nothing goes unobserved.

Chapter 1: The Discovery of the Vera Rubin Observatory

The discovery of the Rubin Observatory was, uniquely among the observatories in this series, a decade of paperwork. The telescope grew out of a shift in how astronomers thought about their own discipline. By the 1990s, the biggest questions in cosmology - how galaxies cluster, how the universe expands, what most of the universe is

made of - were no longer limited by light-gathering power but by sample size. The decisive data would come not from staring longer at famous objects but from measuring faint statistics over vast areas of sky, which meant surveying, and surveying digitally, at a scale no existing instrument could manage. A 1990s generation of sky surveys, led by the Sloan Digital Sky Survey, proved the point: a purpose-built survey telescope, an enormous digital camera, and an open data archive produced a decade of discoveries that changed the textbooks.

The idea that would become Rubin Observatory was voiced most influentially by the astronomer J. Anthony Tyson, who argued for a new telescope built around a wide field of view, a very large detector, and a deliberate obsession with system calibration - an instrument that would image the entire visible sky repeatedly rather than study selected targets. Through the 2000s, the concept circulated under the name Large Synoptic Survey Telescope, LSST: large, synoptic meaning a view of the whole at once, and survey. It gathered institutional momentum through reports and community workshops, and in 2010 the United States decadal survey of astronomy ranked it the top priority for ground-based construction - the single most persuasive endorsement American astronomy can give a project. With that ranking came the political engineering: a partnership between the National Science Foundation, which builds and operates telescopes, and the Department of Energy, which builds detectors and data systems for particle physics, an arrangement that gave the project two masters, two funding lines, and a culture of collaboration between two scientific communities that had rarely built anything together.

The next ingredient was a mountain. A survey telescope needs the darkest, steadiest, driest sky obtainable, and in the early 2000s the project surveyed the Americas for a site before settling on Cerro Pachon, an established observatory ridge in the Chilean Andes already hosting the Gemini South telescope and a southern research telescope run by a consortium. Pachon offered the dark skies of northern Chile, high altitude, and - unusual for a mountaintop observatory - neighbors, with power, roads, and staff already in place. The specific site is a peak beside the ridge, nearly nine thousand feet up, overlooking a landscape of dry hills that has become the densest concentration of major telescopes on Earth. The funding story deserves a paragraph of its own, because it was as inventive as the engineering. Private donors organized under a nonprofit corporation raised early money when the government's commitment was still pending, paying for everything from mirror casting to the salaries of young scientists who would spend their careers on the project; the agencies took over the capital construction under a cooperative agreement, and the finished observatory operates as a federal facility jointly sponsored by the two agencies. That hybrid structure - philanthropy paving the runway, government funding the flight - became a template for the era's biggest science projects, and the private supporters, among them some of the software industry's best-known names, were thanked publicly and early: the telescope itself carries the name of one such benefactor pair. Ground was broken in April 2015, and the construction years that followed read like a slow-motion assembly of a giant: the telescope mount, the dome, the summit facility with its enormous clean rooms, and, arriving from abroad in sequence, the mirrors and the camera.

The heart of the telescope was cast in Tucson, Arizona, in a laboratory that specializes in the impossible. The University of Arizona's mirror lab spins molten glass in a rotating furnace beneath the stadium stands, and for this project it attempted something no one had made before: a single 8.4-meter piece of glass carrying two optical surfaces - a deep primary mirror on one face and a convex tertiary mirror on the other, formed into the same monolith so that light entering the wide field could be folded twice and delivered to the camera with the least possible number of reflections. The casting succeeded, and after years of grinding and polishing the combined mirror - wider than a two-lane road - was completed, coated with silver, and shipped to Chile. In the first week of October 2024 it was lowered into the telescope and mounted, one of the final great strokes of the assembly.

The camera took even longer. Under Department of Energy leadership at the SLAC National Accelerator Laboratory in California, engineers spent two decades designing and building a 3,200-megapixel detector array - the largest digital camera ever constructed for astronomy - with custom silicon sensors developed with Brookhaven National Laboratory, a focal plane flat to a fraction of the width of a human hair, and front optics crowned by the largest lens of its kind ever made, more than a meter and a half across. The completed camera, the size of a small car and weighing about three tons, was finished in April 2024, shipped south, and raised into the telescope's cassegrain cage in early 2025. The last years of assembly fell into a rhythm the project's chroniclers have called the great stacking, because the mountain received its pieces in an order that read like a countdown. The secondary mirror, three and a half meters of polished glass in its own cell, went up in mid-2024; the giant combined primary followed in October, lowered into the telescope's heart; the camera, having crossed an ocean and a continent in its climate-controlled cradle, swung into the cage above the mirror in early 2025, and the first light commissioning nights followed within months, exercised by the smaller commissioning camera in the preceding season before the great sensor array arrived. Each piece had been tested for years before it touched

the mountain, and the assemblage - mirror, camera, telescope, pipelines - was then tested again as a system, until the whole machine behaved like one instrument rather than a committee. On June 23, 2025, the observatory released its first images to the world - wide, deep fields streaked with thousands of asteroid trails - and in late June 2026 the ten-year survey formally began. The long paperwork of the 1990s had become a working machine.